

WASP-10b

R-2311

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_110919 · created 2026-08-18T00:04:07+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2455823.81334 +/- 0.00054 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1694 +/- 0.0094
Transit depth (Rp/Rs)^2	2.87 +/- 0.32 [%]
Orbital Inclination (inc)	89.61 +/- 0.49
Airmass coefficient 1 (a1)	1.1018 +/- 0.0092
Airmass coefficient 2 (a2)	-0.084 +/- 0.031
Scatter in the residuals of the lightcurve fit is	0.83 %
Best Comparison Star	#3 - [347, 129]
Optimal Aperture	3.82
Optimal Annulus	8.95
Transit Duration (day)	0.09261 +/- 0.00075

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1694 ± 0.0094 vs 0.1592 (deviation 1.05σ)
Duration fit vs published	0.09261 d vs 0.0946 d (deviation 2.55σ)
Mid-time O–C	3.0 min
Depth SNR	17.4
Residual scatter	0.83 %

Light curve

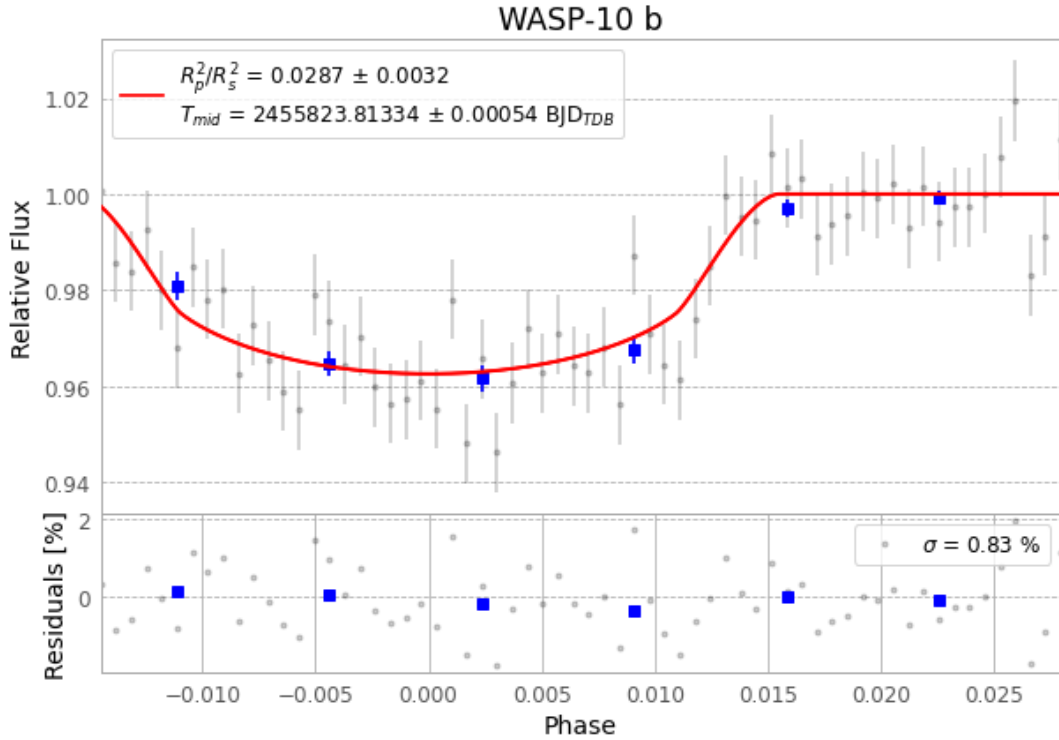


Figure 1 — FinalLightCurve_WASP-10 b_2011-09-18.png (SHA-256 a336b9b2932c009b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [351, 228], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 6f74b11df62622d7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

64 FITS frames (42 MB), WASP-10110919061812.FITS → WASP-10110919092713.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-110919.log (dfed3406525b...), FinalLightCurve_WASP-10 b_2011-09-18.png (a336b9b2932c...), FinalLightCurve_WASP-10 b_2011-09-18.csv (943994aaafd9...)

Compute resources

Wall time 148.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-18 00:04Z.